

TestFlow

Tenant UI/UX Reference for Handoff

Purpose

TestFlow is a multi-tenant B2B SaaS product for electrical substation commissioning. Each client company gets an isolated workspace at `company.testflow.io`. This document describes the tenant-facing UI/UX as it exists in the current codebase, organized by role, so a designer or external collaborator can understand the product without reading source code. It covers screens, navigation, key workflows, and shared visual/interaction conventions. It does not cover the public marketing site or the mobile app (Expo/React Native, ENGINEER-only, same backend).

1 Product Shell & Shared UI

1.1 Tech / Design System

React 18 SPA (Vite, no SSR) styled with **shadcn/ui** on **Tailwind CSS v3**. Motion via **framer-motion** (staggered card fade/slide-ins, hover lifts). Charts via **recharts** (bar/pie). Icons via **lucide-react**. Data layer: TanStack Query v5, with Supabase Realtime subscriptions (debounced invalidation) and a polling fallback when realtime is disabled.

1.2 Host-Based Routing

The same codebase serves three distinct experiences based on hostname:

- `admin.optimustesting.com` — Platform Admin panel (internal, password-gated, no tenant auth)
- `optimustesting.com / www / localhost` — public marketing site
- `<anything>.optimustesting.com` — tenant workspace (this document’s main subject)

1.3 Navigation Shell (DashboardLayout)

Every authenticated tenant screen shares one shell:

- **Fixed left sidebar** (240px): logo block (“TestFlow / Grid Control”), a “Workspace” label showing the user’s role as a portal name (e.g. “GM Portal”, “Manager Portal” for SUPERVISOR), a role-specific nav list, and a bottom block with avatar (colored initials), name, a small role pill, theme toggle, notification bell, and sign-out.
- **Top bar**: sticky, frosted-glass, shows the page title only.
- **Below top bar**: a *Trial Banner* (shown when in trial or subscription is past-due/paused/expired/cancelled) and a *Realtime Status Banner* (shown when the realtime connection has degraded, prompting a page refresh).
- **Content area**: page-specific.

Role pill colors: SUPERADMIN purple, GM blue, SUPERVISOR amber (labeled “manager”), ENGINEER green.

1.4 Status Badge Convention

A single reusable pill component (`StatusBadge`) renders every status value across the product consistently: a small glowing dot + uppercase mono label inside a tinted rounded pill.

Status	Color
DRAFT / UNASSIGNED / CLOSED	slate/gray
ASSIGNED	blue
IN_PROGRESS	cyan
SUBMITTED	violet
APPROVED	emerald green
REWORK	red
ACTIVE (project)	amber

This same badge is reused for project status, equipment status, and test-task status everywhere in the app.

1.5 Auth Screen (/auth)

Two-column layout on desktop (hero panel + form panel), single column on mobile.

- **Hero panel** (left, desktop only): logo, a “Live · Multi-tenant SaaS” pill, headline (“From test sheet to energized bus.”), a 4-item feature list, and 3 stat tiles (Equipment Types: 8, Test Templates: 46, Role Levels: 4).
- **Form panel**: company name (if resolved from subdomain) + “Sign in”, email + password fields (password visibility toggle), “Forgot?” link opening a reset-email dialog, a primary Sign In button, a divider, and a “Sign in with Google” button (disabled per-tenant via a company setting).
- **Special states** replace the form: *Awaiting approval* (Google sign-in pending admin role assignment), *Account deactivated*, *Wrong workspace* (account belongs to a different tenant subdomain).
- **Password reset mode** (`?reset=true`): a minimal centered card with new password + confirm fields.
- No public sign-up exists anywhere — all accounts are admin-provisioned.

1.6 Profile Page (/profile, all roles)

Avatar + name/email header, an “Account Information” card (read-only email, editable display name), and a “Change Password” card (new + confirm password).

2 Role: SUPERADMIN

2.1 Landing: Superadmin Dashboard (/superadmin)

- **Stat cards** (top row): Total Users, Active Users, Active Projects.
- **System Health card**: overall status (Healthy / Degraded / Checking...) with a manual refresh button; auto-refreshes every 30s.
- **Service Status card**: live per-service health rows (Database, Auth, and three Edge Functions: `create-user`, `generate-report`, `platform-admin-data`) each showing a status icon, latency in ms (yellow above 1000ms), or an error detail string.

- **Google Sign-In Approvals card:** queue of users who authenticated via Google OAuth and need a role assigned before they can use the app; per-row role dropdown (Engineer / Supervisor / GM) plus Approve / Reject buttons.
- **User Management card:** embeds the shared `UserManagementTable` (see below).
- **Audit Log card:** embeds `AuditLogViewer`, a company-wide append-only change history (feature-flagged via `audit_log_viewer`).

2.2 Navigation

Dashboard, User Management (same route as Dashboard), Projects (routes to the GM dashboard — SUPERADMIN can see/manage all projects), Reports, Billing, Profile.

2.3 User Management Table (shared component, SUPERADMIN-only surface)

- Toolbar: search (name/email), role filter, status filter, “Bulk Invite” button, “Invite User” button.
- Table columns: Name, Email, Role (colored badge), Status (toggle switch + Active/Inactive label), Actions (Edit Role, Offboard — amber icon, transfers work then deactivates; Delete — destructive, permanent).
- Dialogs: **Invite User** (single), **Bulk Invite** (paste/import list, 800ms delay between sequential invites, stops on rate limit), **Edit Role**, **Offboard** (reassign open work to another user), confirmation alert dialogs for deactivate and delete.
- Self-protection: cannot deactivate or delete your own account.

2.4 Billing Settings (/settings/billing, SUPERADMIN-only)

Plan name/description card with subscription action buttons (upgrade/downgrade/cancel via `SubscriptionActions`), a Usage card with progress bars for Users and Active Projects against plan limits plus an AI-reports-this-month counter, and an Invoices card (placeholder, “coming soon”).

3 Role: GM (General Manager)

3.1 Landing: GM Dashboard (/gm)

- **5 stat cards:** Total Projects, Active, Completed, Assigned, Unassigned — animated counters, hover-lift.
- **“New Test Plan” button** (top right) → project creation wizard.
- **Two charts** (shown once projects exist): a bar chart of project status breakdown (Draft/Approved/Active/Closed) and a donut chart of Assigned vs. Unassigned.
- **Projects list card:** search box, an “All / Assigned / Unassigned” tab filter, paginated list (20 per page, “Show more” button) of project rows — each shows project number, site name/address, client, start date, assigned supervisor (if any), an Assign/Reassign button, an “Overdue” badge when past end date and not closed, and the status badge. Clicking a row opens Project Detail.
- Empty states differ for “no projects yet” (with a create-first-project CTA) vs. “no results for your filter/search”.
- Live-updates via Supabase Realtime (debounced 800ms) with polling fallback.

3.2 Navigation

Projects, New Project, Reports, Profile.

3.3 New Project Wizard (/projects/new, GM/SUPERADMIN)

A 4-step wizard with a progress-stepper header (numbered circles + connecting bar, checkmark on completed steps):

1. **Basic Info** — project number, site name, site address, client, start/end date.
2. **Equipment Scope** — **ScopeManager**: define equipment types + quantities (types include Power Transformer, SF6 Breaker, VCB, CT, CVT, LA, Isolator, Earth Pit, VT).
3. **Testing Scope** — **TestingScopeSelector**: choose which test templates apply per equipment type.
4. **Review** — read-only summary of all three prior steps (project info definition list, equipment scope table with running total, per-equipment-type testing-scope breakdown showing tests × units = task count) with a final “Create Project” button.

On submit, a plan-limit pre-check RPC runs; if the plan’s active-project cap is hit, an **Upgrade Modal** appears instead (see §6.3) rather than creating the project. New projects are created with status **DRAFT**.

3.4 Project Detail (/projects/:id, all roles, edit actions role-gated)

Shared screen across GM/SUPERADMIN/SUPERVISOR (ENGINEER uses a separate task-execution screen, see §5).

- **Header**: back button, project number, status badge, “Overdue” badge, assigned supervisor name, site name.
- **Header actions** (role-gated): Edit (GM/SUPERADMIN only), Assign/Reassign (opens **AssignProjectDialog**), Download PDF, Export Excel (once equipment exists), AI Report (GM/SUPERADMIN, only when project is **CLOSED**, feature-flagged), and project status transition controls (**ProjectStatusActions**).
- **Tabs**: Overview, Scope, Testing Scope, Activity, and Equipment (only shown once equipment instances exist).
 - *Overview*: a progress card (tests approved / total, percentage bar, breakdown of pending-review / in-progress / not-started counts with colored dots) plus a project-info grid (number, status, site name, client, address, start/end/created dates).
 - *Scope*: read-only table of equipment types and quantities with a total row.
 - *Testing Scope*: accordion per equipment type showing enabled/total test count and equipment quantity; each test row has a checkbox (editable only while project is **DRAFT** or **APPROVED**); “Select All / Deselect All” per type; “Save Testing Scope” and “Generate Equipment & Tasks” buttons (the latter is a one-time, idempotent, server-locked action — disabled once already run).
 - *Activity*: a vertical timeline of audit-log entries (created/updated/status-change events) with actor name and timestamp, colored dot per event severity.
 - *Equipment*: accordion per equipment instance (e.g. PTR-001) showing a small stacked assignment bar (per-engineer color-coded segments) in the header, and inside each accordion a card grid of that unit’s test tasks — each with an engineer-assignment dropdown (color dot per engineer) editable by SUPERVISOR/GM/SUPERADMIN.
- **PDF export dialog**, **Excel export** (dynamically-imported **xlsx**), and an **AI Report dialog** (scrollable markdown viewer with a “Download .md” button) are all launched from this screen.

3.5 Edit Project (/projects/:id/edit, GM/SUPERADMIN)

Not read in detail here, but is the write counterpart to project Overview info (same field set as step 1 of the New Project wizard).

4 Role: SUPERVISOR (labeled “Manager” in the UI)

4.1 Landing: Supervisor Dashboard (/supervisor)

- **4 stat cards:** Assigned Projects, Active Projects, Pending Start (status APPROVED), Pending Review (SUBMITTED test-task count).
- **“Tests Pending Approval” card** — the primary work queue:
 - Each row: a checkbox, test name + code, equipment label + type, a link to the project (jumps to Project Detail → Tests tab), and per-row **Rework** / **Approve** buttons.
 - **Bulk actions:** “Select all” checkbox + count; when items are selected, an “Approve selected (N)” button appears. Bulk approve is guarded server-side against concurrent status changes.
 - **Rework dialog:** opened per-task, requires (optional) free-text reason, submits via a destructive-styled “Send for Rework” button; the reason is later shown to the engineer inline on the test form.
 - Approving a task also rolls the parent equipment-instance status up/down based on remaining sibling tasks.
 - Empty state: green checkmark + “No tests pending review”.
- **“Assigned Projects” card:** list of projects assigned to this supervisor by a GM (project number, site name/address, start date, status badge); empty state explains projects appear once a GM assigns them.
- Live-updates via Realtime with debounced invalidation + polling fallback.

4.2 Navigation

Dashboard, Reports, Profile. (No project-creation or user-management access.)

4.3 Project Detail

Same shared screen as GM (§Project Detail above), minus Edit/AI-report actions, plus the Equipment tab’s per-task engineer-assignment control (SUPERVISOR can (re)assign engineers to tests).

5 Role: ENGINEER

5.1 Landing: Engineer Dashboard (/engineer)

- **3 stat cards:** Assigned Tests, In Progress, Completed.
- **“My Projects” card:** one row per project containing equipment assigned to this engineer — project number + status badge, site name/address, start date, “N equipment · X/Y approved” summary, and an animated per-project completion progress bar. Clicking navigates to the engineer’s task-execution screen for that project.
- Empty state: “No assignments yet — Your manager will assign test scopes to you.”

5.2 Navigation

“My Tasks” (dashboard), Profile only — the narrowest nav of any role.

5.3 Task Execution Screen (/engineer/projects/:id)

This is the most complex screen in the product — a dedicated, form-heavy workspace distinct from the shared Project Detail screen used by other roles.

- **Equipment selector card:** a dropdown of equipment units assigned to this engineer within the project, plus a summary (equipment type badge, total unit count).
- **3-step sub-navigation** (numbered pill tabs), scoped to the selected equipment unit:
 1. **Nameplate Details** — a 2-column field grid generated from a per-equipment-type field schema (label, unit suffix, number/text input); “Save Nameplate” button required before proceeding.
 2. **Testing Parameters** — one continuous scrollable form, one section per assigned test: a numbered header with test name/code, a “Draft” indicator when unsaved local changes exist, and the status badge. Renders either a **V2 dynamic form** (`TestFormV2`, sectioned schema-driven fields) or a legacy V1 fallback (per-field inputs/selects from a JSON Schema, or a single generic “Reading Value” input if no schema is defined). Every test section also has: an Instrument Selector, a Pass/Fail/Marginal select, and a Remarks textarea. A **Rework banner** (orange) surfaces the supervisor’s reason when status is REWORK. A sticky action bar offers “Save All Drafts” and “Submit All”. Submitted/Approved tasks render read-only. In-progress edits autosave to `localStorage` as a draft keyed per task, recovered on reload.
 3. **Review & Submit** — read-only nameplate summary table, a test summary list (per test: pass/fail chip + status badge), a submitted-count line, and a final “Submit Equipment” button.
- Saving a test sets task status to `IN_PROGRESS`; submitting sets `SUBMITTED` and clears the local draft; this also recomputes the parent equipment instance’s aggregate status (e.g. all-approved → `APPROVED`, any-rework → `REWORK`).

6 Cross-Role Elements

6.1 Reports

`/reports` (GM, SUPERVISOR, SUPERADMIN) — a card list of all projects (icon, number, status badge, site name, client, start/end dates, arrow) that opens `/reports/:id`, a project-scoped report detail view (not read in full detail for this document, but structurally parallel to Project Detail’s read-only content).

6.2 Notifications

A bell icon in the sidebar (SUPERVISOR and ENGINEER only) opens a popover: SUPERVISORs see newly SUBMITTED tests across their assigned projects; ENGINEERs see tasks sent back for REWORK. Each row shows a colored status dot, message, project/equipment/test breadcrumb, and relative timestamp. Debounced realtime refresh (1.5s) plus polling fallback.

6.3 Upgrade Modal

A shared dialog (`UpgradeModal`) triggered wherever a plan limit blocks an action (e.g. creating a project over the active-project cap). Shows the current limit/usage and the cheapest plan that would unblock the action, with a “View plans” CTA linking to the marketing site’s pricing section.

6.4 Trial / Billing Banner

Shown across all tenant dashboards when relevant: an amber banner for trial ending within 7 days, a red/destructive banner for trial expired or subscription past_due/paused/expired/cancelled, each with a “Manage billing” button. Fully hidden for active paid subscriptions.

7 Platform Admin Panel (Internal, Not Tenant-Facing)

Hosted at `admin.optimustesting.com`; separate password gate (no Supabase Auth session), all data access proxied through a service-role Edge Function. Included here only for completeness — this is an internal ops tool, not part of the customer-facing product.

- **Login**: single password field.
- **Dashboard tabs**: **Companies** (tenant list/detail, create-tenant flow, activate/deactivate, magic-link generation), **Users** (global cross-tenant user search), **Sales** (internal outreach CRM — 7-stage lead pipeline: New → Contacted → Demo Booked → Pilot → Won/Lost/Parked, with a lead detail drawer and activity log), **Billing** (subscription/order inspection, a scaffold around the Razorpay integration).

8 Design Notes for Whoever Picks This Up

- The current visual language is dark-capable shadcn/ui defaults with a blue primary accent, mono uppercase micro-labels (status badges, workspace labels), subtle grid-texture background, and glow effects on the primary logo mark. There is a light/dark theme toggle (`ThemeToggle`) available in the sidebar.
- There is no distinct visual identity per role beyond the sidebar role pill color and the “Portal” label — all four roles share one component library and one visual system.
- The heaviest UX surface by far is the ENGINEER task-execution screen (§5) — long forms, many field types, autosave, and status-driven read-only states. Any redesign effort should prioritize that flow’s information density and error/rework messaging.
- Empty states and loading skeletons exist consistently across list-type screens (projects, users, equipment, notifications) and are a good pattern to preserve.